doesn't *absorb* the light. Oxygen and nitrogen molecules don't absorb visible light; they only bounce it away in some other direction. Other molecules, though, can gobble up the light. Oxides of nitrogen—produced in automotive engines and in the fires of industry—are a source of the murky brown coloration of smog. Oxides of nitrogen (made from oxygen and nitrogen) *do* absorb light. Absorption, as well as scattering, can color a sky.

OTHER WORLDS, OTHER SKIES: Mercury, the Earth's Moon, and most satellites of the other planets are small worlds; because of their feeble gravities, they are unable to retain their atmospheres—which instead trickle off into space. The near-vacuum of space then reaches the ground. Sunlight strikes their surfaces unimpeded, neither scattered nor absorbed along the way. The skies of these worlds are black, even at noon. This has been witnessed firsthand so far by only 12 humans, the lunar landing crews of *Apollos 11, 12*, and *14–17*.

A full list of the satellites in the Solar System, known as of this writing, is given in the accompanying table. (Nearly half of them were discovered by *Voyager)* All have black skies—except Titan of Saturn and perhaps Triton of Neptune, which are big enough to have atmospheres. And all asteroids as well.

Venus has about 90 times more air than Earth. It isn't mainly oxygen and nitrogen as here—it's carbon dioxide. But carbon dioxide doesn't absorb visible light either. What would the sky look like from the surface of Venus if Venus had no clouds? With so much atmosphere in the way, not only are violet and blue waves scattered, but all the other colors as well—green, yellow, orange, red. The air is so thick, though, that hardly any blue light makes it to the ground; it's scattered back to space by successive bounces higher up. Thus, the light that does reach the ground should be strongly reddened—like an Earth sunset all over the sky. Further, sulfur in the high clouds will stain the sky yellow. Pictures taken by the Soviet *Venera* landers confirm that the skies of Venus are a kind of yellow-orange.